INCH-POUND
MIL-PRF-83383/11
10 August 2007

PERFORMANCE SPECIFICATION SHEET

CIRCUIT BREAKERS, REMOTE CONTROL, AC-DC, THERMAL, TRIP-FREE, SERIES TRIP, SINGLE POLE, AUXILIARY CONTACTS, INTERNAL WIRING CONNECTION FOR MULTIPLE LINE PROTECTION (5 TO 100 AMPERES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-83383.

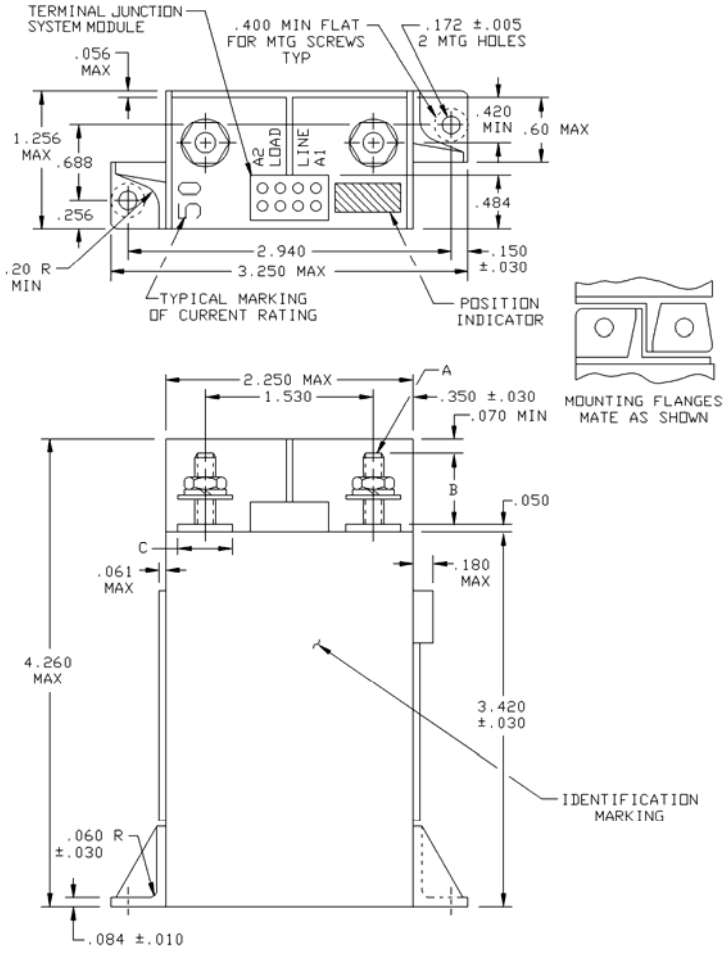


FIGURE 1. Dimensions and configuration.

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Current rating amperes	A	B +.015 -.000	C	Nut 1/	Lock washer 2/	Flat washer 3/
5 to 25	.190-32UNC-2A	.500	.500	AN315-3R	MS35338-43	NAS1149F0332P
35 to 100	.250-28UNF-2A	.610	.610	AN315-4R	MS35338-44	NAS1149F0463P

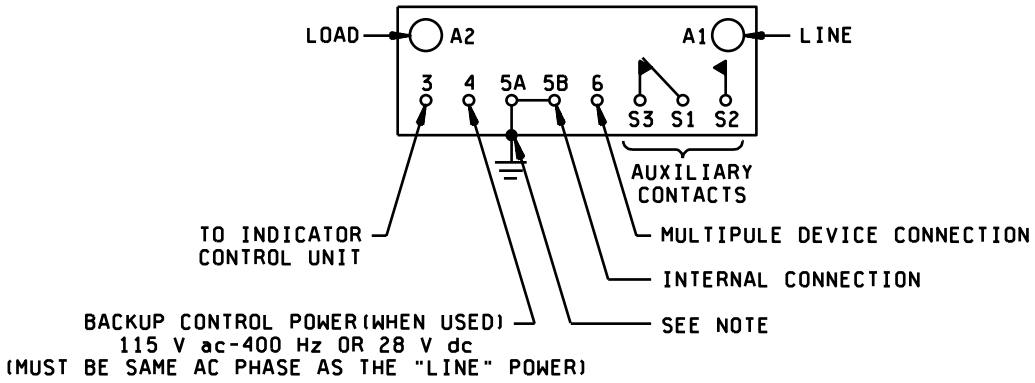
1/ Part number is from National Aerospace Standard NASM315.
2/ Part number is from National Aerospace Standard NASM35338.
3/ Part number is from National Aerospace Standard NAS1149.

Inches	mm	Inches	mm	Inches	mm
.005	0.13	.172	4.37	.60	15.2
.010	0.25	.180	4.57	.610	15.49
.015	0.38	.190	4.83	.688	17.48
.030	0.76	.20	5.1	1.256	31.90
.050	1.27	.250	6.35	1.530	38.86
.056	1.42	.256	6.50	2.250	57.15
.060	1.52	.350	8.89	2.940	74.68
.061	1.55	.400	10.16	3.250	82.55
.070	1.78	.420	10.67	3.420	86.87
.084	2.13	.484	12.29	4.260	108.20
.150	3.81	.500	12.70		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.015 (0.38 mm) for three decimal places and ±.03 (0.76 mm) for two decimal places.
4. Stiffening webs at mounting flanges shown for reference profile as required.
5. Current rating markings may be rotated 90 degrees from the type marking shown in order to accommodate large ampere ratings.

FIGURE 1. Dimensions and configuration – Continued.

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NOTE: Terminals 5A and 5B shall be internally grounded to the mounting flange(s).

FIGURE 2. Schematic.

TABLE I. Dash numbers and characteristics.

Dash number	Current rating (amperes)	Tripping times from -54°C ±5°C to +71°C ±5°C (time in seconds)					
		Percent rated current					
		200		400		1000	
		Min	Max	Min	Max	Min	Max
01	5.0	7	40	1.2	6.4	.3	1.2
02	7.5	11	40	2.4	6.8	.33	1.1
03	10.0	12	42	2.8	8.5	.42	1.05
04	15.0	13	45	1.7	8.3	.35	1.2
05	20.0	14	46	2.9	7.6	.4	1.15
06	25.0	15	50	2.6	8.7	.4	1.3
07	35.0	16	55	2.8	8.3	.35	1.3
08	40.0	16	55	2.9	9.2	.36	1.3
09	50.0	13	55	2.9	10	.4	1.25
10	60.0	13	60	2.6	13	.26	1.8
11	75.0	13	60	2.5	13	.26	1.8
12	80.0	14	60	2.7	12.5	.3	2
13	100.0	17	63	3.5	13	.35	1.9

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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Weight:

5 to 25 amperes: 0.735 pound (333 grams), maximum.

35 to 50 amperes: 0.750 pound (340 grams), maximum.

60 to 100 amperes: 0.766 pound (347 grams), maximum.

Trip indication: See figure 1.

Terminals: See figures 1 and 2.

Auxiliary contact terminals:

Contact capacity at 28 V dc and 115 V ac, 400 Hz shall be 3 amperes resistive, 1.5 amperes inductive, and 0.5 ampere lamp load.

Voltage rating: 28 V dc or 115 V ac, 400 Hz.

Current ratings: See table I.

Tripping times: See table I.

Response time: 25 milliseconds, maximum.

Power requirements and response time at minimum voltage:

Standby current drain and actuation current:

Test voltage: 28 ±0.5 V dc to terminal junction system (TJS) module, then 115 ±2 V ac, 400 Hz to line terminal. Actuating current, 3.5 amperes dc and 11 amperes ac rms maximum.

Maximum operating time at minimum voltage:

Test voltage: 18 V dc then 104 V ac rms, 400 Hz, maximum.

Endurance:

Electrical:

5 to 25 amperes: 50,000 cycles resistive or inductive or motor load or lamp load.

35 to 50 amperes: 50, 000 cycles resistive or inductive or motor load or 25,000 cycles lamp load.

60 to 100 amperes: 50,000 cycles resistive or inductive load or 25,000 cycles motor load.

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Interrupting capacities:

A - 3,600 amperes.

B - 3,600 amperes.

C - 3,600 amperes.

D - 3,600 amperes.

Special busing tests (Pin 3 and Pin 6 internally connected):

Multiple line protection of single pole devices (up to 2 RCCBs):

Verification of the ON- OFF-switching function.

Test voltage: 18 V dc or 104 V ac 400 Hz.

A. Connect I/CU to pin 3 of the first RCCB. Complete control circuit to the remaining RCCBs (connect pin 6 of the first RCCB to pin 3 of the second RCCB). Consequently the I/CU of the first RCCB will simultaneously control the second RCCB. Make sure to connect each RCCB's mounting flange or terminal 5A/5B to GND.
B. Apply test voltage to line terminals (A1) on each RCCB.
C. Manually close the I/CU. All interconnected RCCBs shall turn ON.
D. Manually open the I/CU. All interconnected RCCBs shall turn OFF. Remove all power from the RCCBs.

Verification of the trip-free-function.

Test voltage: 18 V dc or 104 V ac 400 Hz.

Impedance between pin 3 and GND must not exceed 7.5 Ω.

A. Apply test voltage to line terminals (A1) on each RCCB.
B. Manually close I/CU. All interconnected RCCBs shall turn ON.
C. Apply 200 percent overload on the first RCCB main circuit path (A1 to A2). The RCCB shall trip and I/CU shall open, and all interconnected RCCBs shall open. Verify their trip-free function as specified in Section 3.27.
D. Repeat step 2-3 on each RCCB.

Verification of 0 Ω resistive connection between pin 3 and 6.

Resistive connection on terminal junction system module between pin 3 and 6 shall measure 0 Ω.

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Other marking:

Current rating: See figure 1.

Line and load terminals: See figure 1.

TJS module: See figure 3.

Position indication: See figure 1. Indication is provided by a lens over “open” and “closed” on red and green background.

Part number: M83383/11- (dash number from table I) (e.g., M83383/11-01).

Indicator/control unit (I/CU): In order to maintain compatibility between the RCCB and the I/CU, the following must be observed when making an I/CU selection:

a. The I/CU must be a 1/2 ampere rated circuit breaker (SAE-ASS22073-1/2 or SAE-ASS26574-1/2, as specified in SAE-AS58091, or equivalent).

b. The line impedance (terminal 3 to I/CU) plus the I/CU impedance plus I/CU to ground connection.

c. The I/CU must respond to a decreasing current pulse of $I^2t \leq 2$ throughout the temperature range.

CAUTION TO USERS:

Coordination between devices supplied by different manufacturers should be verified by the user.

It is not possible to identify which RCCB trips first because all RCBBs will open after a fault.

Potentially there may be additional electromagnetic interference due to external wiring.

Referenced documents. In addition to MIL-PRF-83383, this document references the following:

NASM315 NASM35338 NAS1149 SAE-AS58091

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Custodians:
Army - CR
Navy - AS
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5925-2007-013)

Review activities:
Army - AT, AV, CR4
Air Force - 99

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